Assignment #2 Swanstrom

Grade: CheckPlus

1. EXPECTED VALUES AND OPTIONS

A stock has a current selling price of $20. The distribution of the price of the stock one year from now is believed to be as follows:

PRICE (x)	PROBABILITY f(x)
20	.05
21	.06
22	.07
23	.08
24	.09
25	.12
26	.15
27	.13
28	.10
29	.10
30	.05
	1.00

An investor holds a one year option (European) on this stock. This enables him to purchase, one year from now, one share of stock at $25 per share. The value of his option will depend upon the price of the stock one year from now. If the price rises to $28, the value of the option (ignoring interest and commissions) is $3. If the price rises to only $23, the option is worthless.

1. Compute the mean and standard deviation of the stock price.
2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?
   b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?
3. Derive the distribution of the option value, ignoring interest and commissions.
4. What is the probability that the option will have positive value?
5. Compute the mean and standard deviation of the value of the option.

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Assignment 2

1.) mean and standard deviation of stock pile
See column C for calculation of mean using weighted average technique $x \cdot f(x)$ to get 25.41 as the mean

x Q1: Your mean is correct, but the standard...

standard deviation was calculated using the variance (column D) and taking the square root of this to get 2.72 as the standard deviation

A.)
2.) $P(25.41 - 2.72 \le x \le 25.41 + 2.72)$
$22.69 \le x \le 28.13$ ✓ Q2a
$P(23 \le x \le 28) = f(23) + f(24) + f(25) + f(26) + f(27) + f(28)$
$= 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10$
$= 0.67$ or 67%

B.) $P(25.41 - 5.44 \le x \le 25.41 + 5.44)$
$P(19.97 \le x \le 30.85)$ ✓ Q2b
$P(20 \le x \le 30) =$ all possibilities included so 100%

3.) See column E & F for calculations & distribution

data (for questions 4 & 5)

X	value	f(x) probability	option value · f(x)	variance
20-25	0	0.47	0	0.86
26	1	0.15	0.15	0.02
27	2	0.13	0.26	0.05
28	3	0.1	0.3	0.26
29	4	0.1	0.4	0.70
30	5	0.05	0.25	0.66
		1	1.36 = mean	2.55

✓ Q3

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Assignment 2 continued

✓ Q4

4.) 1 - 0.47 - 0.53 or 53%

✓ Q5

5.) mean = 1.36 (see calculations in Q3 table)

stand. deviation = 1.6 (see calculations in Q3 table)

$\sqrt{\text{variance}} = \sqrt{2.57} = 1.6$